

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284414

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Sharma; Amit et al.

Data Storage Device and Method for Power Saving in a Variable Host Throughput Environment

Abstract

A data storage device and method for power saving in a variable host throughput environment are provided. In one embodiment, a data storage device is provided comprising a memory and one or more processors. The one or more processors, individually or in combination, are configured to: detect that a data consumption rate of a host is below a threshold; and in response to detecting that the data consumption rate of the host is below the threshold, enter a power-saving mode by: storing, in the non-volatile memory, a context of a pending host command in a command queue; and reducing power to at least one component of the data storage device. Other embodiments are disclosed.

Inventors: Sharma; Amit (Bangalore, IN), Agarwal; Dinesh Kumar (Bangalore, IN), Venugopal; Abhinandan (Mysore, IN)

Applicant: Western Digital Technologies, Inc. (San Jose, CA)

Family ID: 1000007773181

Assignee: Western Digital Technologies, Inc. (San Jose, CA)

Appl. No.: 18/596212

Filed: March 05, 2024

Publication Classification

Int. Cl.: G06F3/06 (20060101)

U.S. Cl.:

CPC G06F3/0625 (20130101); G06F3/0634 (20130101); G06F3/0653 (20130101); G06F3/0679 (20130101);

Background/Summary

BACKGROUND

[0001] Some data storage devices are host-powered and can enter a low-power mode to reduce the amount of consumed host power. For example, a data storage device can enter a low-power mode when no host commands are pending and can exit the low-power mode upon receiving a new host command.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1A is a block diagram of a data storage device of an embodiment.

[0003] FIG. 1B is a block diagram illustrating a storage module of an embodiment.

[0004] FIG. 1C is a block diagram illustrating a hierarchical storage system of an embodiment.

[0005] FIG. 2A is a block diagram illustrating components of the controller of the data storage device illustrated in FIG. 1A according to an embodiment.

[0006] FIG. 2B is a block diagram illustrating components of the data storage device illustrated in FIG. 1A according to an embodiment.

[0007] FIG. 3 is a block diagram of a host and a data storage device of an embodiment.

[0008] FIG. 4 is a block diagram of a vehicle and a solid-state drive (SSD) of an embodiment.

[0009] FIG. 5 is a flow diagram of a host-device handshake of an embodiment.

[0010] FIG. 6 is an illustration of an embodiment in which pending host commands prevent a data storage device from entering a low-power mode.

[0011] FIG. 7 is an illustration of an embodiment in which pending host commands do not prevent a data storage device from entering a low-power mode.

[0012] FIG. 8 is a flow diagram of a host-device handshake of an embodiment that takes place during device initialization.

DETAILED DESCRIPTION

[0013] The following embodiments generally relate to a data storage device and method for power saving in a variable host throughput environment. In one embodiment, a data storage device is provided comprising a memory and one or more processors. The one or more processors, individually or in combination, are configured to: detect that a data consumption rate of a host is below a threshold; and in response to detecting that the data consumption rate of the host is below the threshold, enter a power-saving mode by: storing, in the non-volatile memory, a context of a pending host command in a command queue; and reducing power to at least one component of the data storage device.

[0014] In some embodiments, the one or more processors, individually or in combination, are further configured to exit the power-saving mode by: reading, from the non-volatile memory, the context of the pending host command; and restoring power to the at least one component of the data storage device.

[0015] In some embodiments, the one or more processors, individually or in combination, are further configured to exit the power-saving mode in response to an expiration of a time period.

[0016] In some embodiments, the time period is a function of the data consumption rate of the host and a size of a buffer in the host.

[0017] In some embodiments, the one or more processors, individually or in combination, are further configured to exit the power-saving mode in response to receiving a new host command from the host.

[0018] In some embodiments, the one or more processors, individually or in combination, are

further configured to exit the power-saving mode in response to receiving a command from the host to exit the power-saving mode.

[0019] In some embodiments, the one or more processors, individually or in combination, are further configured to seek approval from the host of an increase in latency in exiting the power-saving mode.

[0020] In some embodiments, the one or more processors, individually or in combination, are further configured to request that the host not send a new host command to the data storage device while the data storage device is in the power-saving mode.

[0021] In some embodiments, the one or more processors, individually or in combination, are further configured to perform a handshake with the host to enable use of the power-saving mode.

[0022] In some embodiments, the non-volatile memory comprises a three-dimensional memory.

[0023] In another embodiment, a method is provided that is performed in a data storage device. The method comprises: determining that there is at least one outstanding host command that has not been completed; and entering a low-power mode even though there is the at least one outstanding host command that has not been completed.

[0024] In some embodiments, entering the low-power mode comprises: storing a state of the at least one outstanding host command; and reducing power to at least one component of the data storage device.

[0025] In some embodiments, the method further comprises exiting the low-power mode.

[0026] In some embodiments, exiting the low-power mode comprises: restoring the state of the at least one outstanding host command; and restoring power to the at least one component of the data storage device.

[0027] In some embodiments, the low-power mode is exited in response to an expiration of a time period, receiving a new host command from the host, and/or receiving a command from the host to exit the low-power mode.

[0028] In some embodiments, the method further comprises receiving consent from the host for an increase in latency in exiting the low-power mode.

[0029] In some embodiments, the method further comprises requesting that the host not send a new host command to the data storage device while the data storage device is in the low-power mode.

[0030] In some embodiments, the method further comprises performing a handshake with the host to enable use of the low-power mode.

[0031] In some embodiments, the host comprises an autonomous vehicle.

[0032] In yet another embodiment, a data storage device is provided comprising: a memory; and means for enter a standby mode in response to detecting that a data consumption rate of a host is below a threshold, wherein the standby mode power is entered even though there is at least one pending host command.

[0033] Other embodiments are possible, and each of the embodiments can be used alone or together in combination. Accordingly, various embodiments will now be described with reference to the attached drawings.

EMBODIMENTS

[0034] The following embodiments relate to a data storage device (DSD). As used herein, a “data storage device” refers to a non-volatile device that stores data. Examples of DSDs include, but are not limited to, hard disk drives (HDDs), solid state drives (SSDs), tape drives, hybrid drives, etc. Details of example DSDs are provided below.

[0035] Examples of data storage devices suitable for use in implementing aspects of these embodiments are shown in FIGS. 1A-1C. It should be noted that these are merely examples and that other implementations can be used. FIG. 1A is a block diagram illustrating the data storage device **100** according to an embodiment. Referring to FIG. 1A, the data storage device **100** in this example includes a controller **102** coupled with a non-volatile memory that may be made up of one or more non-volatile memory die **104**. As used herein, the term die refers to the collection of non-

volatile memory cells, and associated circuitry for managing the physical operation of those non-volatile memory cells, that are formed on a single semiconductor substrate. The controller **102** interfaces with a host system and transmits command sequences for read, program, and erase operations to non-volatile memory die **104**. Also, as used herein, the phrase “in communication with” or “coupled with” could mean directly in communication/coupled with or indirectly in communication/coupled with through one or more components, which may or may not be shown or described herein. The communication/coupling can be wired or wireless.

[0036] The controller **102** (which may be a non-volatile memory controller (e.g., a flash, resistive random-access memory (ReRAM), phase-change memory (PCM), or magnetoresistive random-access memory (MRAM) controller)) can include one or more components, individually or in combination, configured to perform certain functions, including, but not limited to, the functions described herein and illustrated in the flow charts. For example, as shown in FIG. 2A, the controller **102** can comprise one or more processors **138** that are, individually or in combination, configured to perform functions, such as, but not limited to the functions described herein and illustrated in the flow charts, by executing computer-readable program code stored in one or more non-transitory memories **139** inside the controller **102** and/or outside the controller **102** (e.g., in random access memory (RAM) **116** or read-only memory (ROM) **118**). As another example, the one or more components can include circuitry, such as, but not limited to, logic gates, switches, an application specific integrated circuit (ASIC), a programmable logic controller, and an embedded microcontroller.

[0037] In one example embodiment, the non-volatile memory controller **102** is a device that manages data stored on non-volatile memory and communicates with a host, such as a computer or electronic device, with any suitable operating system. The non-volatile memory controller **102** can have various functionality in addition to the specific functionality described herein. For example, the non-volatile memory controller can format the non-volatile memory to ensure the memory is operating properly, map out bad non-volatile memory cells, and allocate spare cells to be substituted for future failed cells. Some part of the spare cells can be used to hold firmware (and/or other metadata used for housekeeping and tracking) to operate the non-volatile memory controller and implement other features. In operation, when a host needs to read data from or write data to the non-volatile memory, it can communicate with the non-volatile memory controller. If the host provides a logical address to which data is to be read/written, the non-volatile memory controller can convert the logical address received from the host to a physical address in the non-volatile memory. The non-volatile memory controller can also perform various memory management functions, such as, but not limited to, wear leveling (distributing writes to avoid wearing out specific blocks of memory that would otherwise be repeatedly written to) and garbage collection (after a block is full, moving only the valid pages of data to a new block, so the full block can be erased and reused).

[0038] Non-volatile memory die **104** may include any suitable non-volatile storage medium, including resistive random-access memory (ReRAM), magnetoresistive random-access memory (MRAM), phase-change memory (PCM), NAND flash memory cells and/or NOR flash memory cells. The memory cells can take the form of solid-state (e.g., flash) memory cells and can be one-time programmable, few-time programmable, or many-time programmable. The memory cells can also be single-level cells (SLC), multiple-level cells (MLC) (e.g., dual-level cells, triple-level cells (TLC), quad-level cells (QLC), etc.) or use other memory cell level technologies, now known or later developed. Also, the memory cells can be fabricated in a two-dimensional or three-dimensional fashion.

[0039] The interface between controller **102** and non-volatile memory die **104** may be any suitable flash interface, such as Toggle Mode **200**, **400**, or **800**. In one embodiment, the data storage device **100** may be a card-based system, such as a secure digital (SD) or a micro secure digital (micro-SD) card. In an alternate embodiment, the data storage device **100** may be part of an embedded data

storage device.

[0040] Although, in the example illustrated in FIG. 1A, the data storage device **100** (sometimes referred to herein as a storage module) includes a single channel between controller **102** and non-volatile memory die **104**, the subject matter described herein is not limited to having a single memory channel. For example, in some architectures (such as the ones shown in FIGS. 1B and 1C), two, four, eight or more memory channels may exist between the controller and the memory device, depending on controller capabilities. In any of the embodiments described herein, more than a single channel may exist between the controller and the memory die, even if a single channel is shown in the drawings.

[0041] FIG. 1B illustrates a storage module **200** that includes plural non-volatile data storage devices **100**. As such, storage module **200** may include a storage controller **202** that interfaces with a host and with data storage device **204**, which includes a plurality of data storage devices **100**. The interface between storage controller **202** and data storage devices **100** may be a bus interface, such as a serial advanced technology attachment (SATA), peripheral component interconnect express (PCIe) interface, double-data-rate (DDR) interface, or serial attached small scale compute interface (SAS/SCSI). Storage module **200**, in one embodiment, may be a solid-state drive (SSD), or non-volatile dual in-line memory module (NVDIMM), such as found in server PC or portable computing devices, such as laptop computers, and tablet computers.

[0042] FIG. 1C is a block diagram illustrating a hierarchical storage system. A hierarchical storage system **250** includes a plurality of storage controllers **202**, each of which controls a respective data storage device **204**. Host systems **252** may access memories within the storage system **250** via a bus interface. In one embodiment, the bus interface may be a Non-Volatile Memory Express (NVMe) or Fibre Channel over Ethernet (FCOE) interface. In one embodiment, the system illustrated in FIG. 1C may be a rack mountable mass storage system that is accessible by multiple host computers, such as would be found in a data center or other location where mass storage is needed.

[0043] Referring again to FIG. 2A, the controller **102** in this example also includes a front-end module **108** that interfaces with a host, a back-end module **110** that interfaces with the one or more non-volatile memory die **104**, and various other components or modules, such as, but not limited to, a buffer manager/bus controller module that manage buffers in RAM **116** and controls the internal bus arbitration of controller **102**. A module can include one or more processors or components, as discussed above. The ROM **118** can store system boot code. Although illustrated in FIG. 2A as located separately from the controller **102**, in other embodiments one or both of the RAM **116** and ROM **118** may be located within the controller **102**. In yet other embodiments, portions of RAM **116** and ROM **118** may be located both within the controller **102** and outside the controller **102**.

[0044] Front-end module **108** includes a host interface **120** and a physical layer interface (PHY) **122** that provide the electrical interface with the host or next level storage controller. The choice of the type of host interface **120** can depend on the type of memory being used. Examples of host interfaces **120** include, but are not limited to, SATA, SATA Express, serially attached small computer system interface (SAS), Fibre Channel, universal serial bus (USB), PCIe, and NVMe. The host interface **120** typically facilitates transfer for data, control signals, and timing signals.

[0045] Back-end module **110** includes an error correction code (ECC) engine **124** that encodes the data bytes received from the host, and decodes and error corrects the data bytes read from the non-volatile memory. A command sequencer **126** generates command sequences, such as program and erase command sequences, to be transmitted to non-volatile memory die **104**. A RAID (Redundant Array of Independent Drives) module **128** manages generation of RAID parity and recovery of failed data. The RAID parity may be used as an additional level of integrity protection for the data being written into the memory device **104**. In some cases, the RAID module **128** may be a part of the ECC engine **124**. A memory interface **130** provides the command sequences to non-volatile

memory die **104** and receives status information from non-volatile memory die **104**. In one embodiment, memory interface **130** may be a double data rate (DDR) interface, such as a Toggle Mode **200**, **400**, or **800** interface. The controller **102** in this example also comprises a media management layer **137** and a flash control layer **132**, which controls the overall operation of back-end module **110**.

[0046] The data storage device **100** also includes other discrete components **140**, such as external electrical interfaces, external RAM, resistors, capacitors, or other components that may interface with controller **102**. In alternative embodiments, one or more of the physical layer interface **122**, RAID module **128**, media management layer **138** and buffer management/bus controller **114** are optional components that are not necessary in the controller **102**.

[0047] FIG. 2B is a block diagram illustrating components of non-volatile memory die **104** in more detail. Non-volatile memory die **104** includes peripheral circuitry **141** and non-volatile memory array **142**. Non-volatile memory array **142** includes the non-volatile memory cells used to store data. The non-volatile memory cells may be any suitable non-volatile memory cells, including ReRAM, MRAM, PCM, NAND flash memory cells and/or NOR flash memory cells in a two-dimensional and/or three-dimensional configuration. Non-volatile memory die **104** further includes a data cache **156** that caches data. The peripheral circuitry **141** in this example includes a state machine **152** that provides status information to the controller **102**. The peripheral circuitry **141** can also comprise one or more components that are, individually or in combination, configured to perform certain functions, including, but not limited to, the functions described herein and illustrated in the flow charts. For example, as shown in FIG. 2B, the memory die **104** can comprise one or more processors **168** that are, individually or in combination, configured to execute computer-readable program code stored in one or more non-transitory memories **169**, stored in the memory array **142**, or stored outside the memory die **104**. As another example, the one or more components can include circuitry, such as, but not limited to, logic gates, switches, an application specific integrated circuit (ASIC), a programmable logic controller, and an embedded microcontroller.

[0048] In addition to or instead of the one or more processors **138** (or, more generally, components) in the controller **102** and the one or more processors **168** (or, more generally, components) in the memory die **104**, the data storage device **100** can comprise another set of one or more processors (or, more generally, components). In general, wherever they are located and however many there are, one or more processors (or, more generally, components) in the data storage device **100** can be, individually or in combination, configured to perform various functions, including, but not limited to, the functions described herein and illustrated in the flow charts. For example, the one or more processors (or components) can be in the controller **102**, memory device **104**, and/or other location in the data storage device **100**. Also, different functions can be performed using different processors (or components) or combinations of processors (or components). Further, means for performing a function can be implemented with a controller comprising one or more components (e.g., processors or the other components described above).

[0049] Returning again to FIG. 2A, the flash control layer **132** (which will be referred to herein as the flash translation layer (FTL) handles flash errors and interfaces with the host. In particular, the FTL, which may be an algorithm in firmware, is responsible for the internals of memory management and translates writes from the host into writes to the memory **104**. The FTL may be needed because the memory **104** may have limited endurance, may be written in only multiples of pages, and/or may not be written unless it is erased as a block. The FTL understands these potential limitations of the memory **104**, which may not be visible to the host. Accordingly, the FTL attempts to translate the writes from host into writes into the memory **104**.

[0050] The FTL may include a logical-to-physical address (L2P) map (sometimes referred to herein as a table or data structure) and allotted cache memory. In this way, the FTL translates logical block addresses (“LBAs”) from the host to physical addresses in the memory **104**. The FTL can include

other features, such as, but not limited to, power-off recovery (so that the data structures of the FTL can be recovered in the event of a sudden power loss) and wear leveling (so that the wear across memory blocks is even to prevent certain blocks from excessive wear, which would result in a greater chance of failure).

[0051] Turning again to the drawings, FIG. 3 is a block diagram of a host **300** and data storage device **100** of an embodiment. The host **300** can take any suitable form, including, but not limited to, a computer, a mobile phone, a tablet, a wearable device, a digital video recorder, a surveillance system, etc. The host **300** in this embodiment (here, a computing device) comprises one or more processors **330** and one or more memories **340**. In one embodiment, computer-readable program code stored in the one or more memories **340** configures the one or more processors **330** to perform the acts described herein as being performed by the host **300**. So, actions performed by the host **300** are sometimes referred to herein as being performed by an application (computer-readable program code) run on the host **300**. For example, the host **300** can be configured to send data (e.g., initially stored in the host's memory **340**) to the data storage device **100** for storage in the data storage device's memory **104**.

[0052] Low power consumption is often desired by hosts ranging from portable data storage devices to data storage devices in data centers. Various host protocols can specify power levels for a data storage device, and each power level can consume power within specified limits. In general, higher power levels/modes are more power efficient. The following is an example classification system:

TABLE-US-00001 Power Mode/Level Brief Description
PM0 (Active) Data storage device is active and performing NAND operations
PM1 (Device Idle) Data storage device is idle, and host commands are pending. The data storage device is allowed to perform background operations. If no host commands are pending, a timeframe is given to the data storage device to perform background operations
PM2 (Standby) Data storage device is in standby mode with no active host commands.

[0053] Of the example power levels specified above, the PM2 state would consume the least power. To enter this mode, the data storage device should not have any pending host commands. So, when there are no pending host commands, a data storage device can implement higher power modes by cutting power to various components (e.g., RAM and other hardware components). With each power mode, the data storage device can be configured to honor exit latency (e.g., a timeframe in which the data storage device should be able to accept/serve a new host command). For example, if the data storage device is in the PM2 state and a new host command arrives, the data storage device should be ready to serve this command in a specified timeframe (e.g., in a few milliseconds). However, there are situations where there can be drop in the rate at which the host is able to consume data. For example, an autonomous vehicle can generate a very large amount of data that is stored locally and then transferred to cloud/servers for training/investigation/decision-making purposes. Because the data generated is so large, the data is transferred from the vehicle (host) to the cloud/servers using wireless communication can take a relatively-long time. An example of this environment is shown in FIG. 4.

[0054] As shown in FIG. 4, in this example environment, the host **300** (here, an autonomous vehicle) is in communication with the data storage device **100** (here, a solid-state drive (SSD)). The host **300** has data transfer logic **390**, which can be implemented by the one or more processors **330** in the host **300** executing computer-readable program instructions stored in the one or more memories **340** in the host **300**. The data transfer logic **390** is configured to control the transfer of data in to and/or out of the host **300**. In operation, the host **300** generates data and stores it in the memory of the data storage device **100**. At some point, the host **300** reads the data from the data storage device **100** and transfers the data to a network, "the cloud," a server, etc. (e.g., via a wireless transceiver/antenna **385**, a wired connector, etc.). However, there can be situations where network connectivity is low, in which case data consumption at the host **300** would be reduced. From the data storage device's perspective, the data storage device **100** would observe multiple host

commands pending in the queue **400** (which can be in the data storage device **100**, the host **300**, and/or another location), but the rate at which the host **300** is consuming data (or command completion) would be low. Because there are commands pending in the queue **400**, the data storage device **100** cannot go to PM2 mode under the example power scheme discussed above, as the primary condition for the data storage device **100** to go to deep power-save mode is not having any pending host commands. So, even though there is minimal or no response required from the data storage device **100**, the data storage device **100** cannot enter standby mode because of the presence of the pending host commands in the queue **400**.

[0055] The following embodiments can be used to address this problem by allowing the host **300** and the data storage device **100** to work together to enable the data storage device **100** to enter a power-saving mode. As used herein, a power-saving mode (which is sometimes referred to herein as a low-power mode or a standby mode) can generally refer to a mode that consumes relatively-less power than another mode (e.g., a “normal” mode of operation or any other mode that consumes relatively-more power). In the power-saving mode, power can be removed or reduced from one or more hardware components in the data storage device **100** (e.g., hardware cores, volatile memories, a portion of the non-volatile memory **104**, latching logic, a physical layer interface, etc.).

[0056] A power-saving mode can be configured to allow the data storage device **100** to save power without impacting performance. In one embodiment, the controller **102** of the data storage device **100** is configured to detect situations where even though host commands are pending in the data storage device **100**, the rate at which the host **300** is consuming/providing data is below a threshold (e.g., a configurable or pre-determined threshold). In these situations, the controller **102** can save the pending host command context or state and enter the power-saving mode even though there are host commands pending in the data storage device **100**. As used herein, a command context can refer to various data generated or used by a command in its execution. For example, command number, LBAs information, user data stored in a write cache buffer, management data (e.g., logical-to-physical address mapping data) stored in volatile memory, data latched in latching logic, etc. can be written (flushed) to the non-volatile memory **104** and/or stored in volatile memory (if any) that is powered during the power-saving mode.

[0057] When the data storage device **100** wakes up, the controller **102** can load back the host command contexts and resume host command execution. Further, the controller **102** of the data storage device **100** can collaborate with the host **300** so that the host **300** would not queue up more commands for certain time, as queueing up a new command can take the data storage device **100** out of the power-saving mode (which may not be a problem for the host **300**, as the host **300** may not be consuming data at a lower rate). As the data storage device **100** may need additional time to load the command context after exiting power-saving mode, the host **300** can acknowledge that this extra latency is acceptable.

[0058] These embodiments can be implemented in any suitable way. The following paragraphs provide examples of such implementations. It should be noted that these are merely examples and that other implementations can be used.

[0059] In one example implementation, the controller **102** is configured to detect situations where the host's data consumption speed is reduced (e.g., from 100 megabytes per second (MBps) to one MBps) or stalls (e.g., for some period of time). In such situation as the host **300** does not need data for some time (or at lower rate), the controller **102** can cause the data storage device **100** to enter a low-power mode to conserve power. However, as noted above, if there are pending host commands, the data storage device **100** may be prevented from entering the low-power mode. Also, even if the data storage device **100** enters the low-power mode, the host **300** can queue one or more new commands, which would cause the data storage device **100** to exit the low-power mode. Further, even when in low-power mode, certain volatile memory (e.g., RAM) components/clocks may need power to keep functioning. So, in this embodiment, the controller **102** can save the

context of the pending host commands in such volatile memory and/or the non-volatile memory **104**. When the data storage device **100** exits the low-power mode, the controller **102** can fetch the command information from its storage location and resume command execution. In an alternate implementation, the components in the data storage device **100** used for the host command queue can be powered on during the low-power mode even though they may consume more power compared to the “standard” low-power mode.

[0060] As mentioned above, there can be issues originating from host side. For example, the host **300** sending command(s) to the data storage device **100** while it is in low-power mode can cause the data storage device **100** to exit the low-power mode. The data storage device **100** typically does not have control over this, as it typically cannot cause the host **300** to delay sending commands to the data storage device **100**. Also, the host **300** can experience extra latencies when the data storage device **100** exits the low-power mode, as the controller **102** of the data storage device **100** needs time to load the host command contexts from their storage locations.

[0061] To address these issues, a host-device “handshake” can be used. More specifically, in the situation where the controller **102** of the data storage device **100** detects that there is a possibility of entering the low-power mode, the controller **102** can communicate with the host **300** to make one or more requests. For example, the controller **102** can request that the host **300** not send more host commands to the data storage device **100** for a certain amount of time. This should not be an issue for the host **300**, since host itself is slower at consuming data (which is the first reason the device **100** is entering the power saving mode). In response to the request, the host **300** can check its internal buffers (e.g., in the one or more memories **340**) to see how long the buffered data can satisfy the needs of the host **300** before the host **300** needs to send a new command to the data storage device **100**.

[0062] Regarding the additional latency from exiting the low-power mode, the controller **102** can ask the host **300** if such additional latency is acceptable. In many situations, it is expected that the host **300** will accept the additional latency, as when the data storage device **100** does into low-power mode, it may already be determined that device speed will not impact the performance of the host **300**.

[0063] FIG. 5 is a flow diagram of a host-device handshake of an embodiment. As shown in FIG. 5, the data storage device **100** detects a low data consumption rate from the host **300** (act **501**) and asks the host **300** if the host **300** can go to an aggressive standby mode (act **502**). In this example, the host **300** has a 100 MB buffer, the data consumption is one MB per second (MBps), and the host **300** notes not to send commands to the data storage device **100** for 100 seconds. The host **300** then gives an affirmative response to the data storage device **100** and can also give an indication to be in standby for 100 seconds (act **504**). In response, the data storage device **100** enters standby mode even when host commands are pending (act **505**). The data storage device **100** wakes up after a given time, or upon receiving a new host command, or upon receiving a host indication to exit standby mode (act **506**).

[0064] This embodiment can help address the problem noted above when a relative-large amount of data is wirelessly transferred from the host **300** (an autonomous vehicle) to a network. For example, assume the host **300** has a 100 MB internal buffer from which transfers data to a wireless network. If the wireless network transfer speed drops from 10 MBps to 1 MBps, the host **300** has enough data for 100 seconds. The host **300** can inform the data storage device **100** that it can go into low-power mode for 100 seconds. Alternatively, the host **300** can send an indication/command for the data storage device **100** to exit the low-power mode.

[0065] FIGS. 6 and 7 illustrate an advantage of this embodiment. FIG. 6 shows that, without using this embodiment, pending host commands can prevent the data storage device **100** from entering a low-power mode. FIG. 7 shows that, with this embodiment, pending host commands do not prevent a data storage device from entering a low-power mode.

[0066] Another approach to achieve this communication can be a host-device handshake during

device initialization. Once the host **300** knows about this new standby mode, the host **300** can enable or disable this feature. If this mode is enabled, the data storage device **100** is allowed to go to the new standby mode even if there are commands pending. In this example handshake, the host **300** is agreeing to some additional delay when the data storage device comes out of the standby mode.

[0067] FIG. **8** is a flow diagram of a host-device handshake of an embodiment that takes place during device initialization. As shown in FIG. **8**, the host **300** asks the data storage device **100** if it supports the newer standby mode (act **801**), and the data storage device **100** sends an acknowledgement (act **802**). The host **300** enables this move to give permission to the data storage device **100** to go to the newer standby mode (act **803**). The data storage device **100** then detects a low data consumption rate from the host **300** (act **804**) and enters standby mode even when host commands are pending (act **805**). The data storage device **100** wakes up after a given time, a new host command, or a host indication to exit this mode (act **807**).

[0068] Finally, as mentioned above, any suitable type of memory can be used. Semiconductor memory devices include volatile memory devices, such as dynamic random access memory (“DRAM”) or static random access memory (“SRAM”) devices, non-volatile memory devices, such as resistive random access memory (“ReRAM”), electrically erasable programmable read only memory (“EEPROM”), flash memory (which can also be considered a subset of EEPROM), ferroelectric random access memory (“FRAM”), and magnetoresistive random access memory (“MRAM”), and other semiconductor elements capable of storing information. Each type of memory device may have different configurations. For example, flash memory devices may be configured in a NAND or a NOR configuration.

[0069] The memory devices can be formed from passive and/or active elements, in any combinations. By way of non-limiting example, passive semiconductor memory elements include ReRAM device elements, which in some embodiments include a resistivity switching storage element, such as an anti-fuse, phase change material, etc., and optionally a steering element, such as a diode, etc. Further by way of non-limiting example, active semiconductor memory elements include EEPROM and flash memory device elements, which in some embodiments include elements containing a charge storage region, such as a floating gate, conductive nanoparticles, or a charge storage dielectric material.

[0070] Multiple memory elements may be configured so that they are connected in series or so that each element is individually accessible. By way of non-limiting example, flash memory devices in a NAND configuration (NAND memory) typically contain memory elements connected in series. A NAND memory array may be configured so that the array is composed of multiple strings of memory in which a string is composed of multiple memory elements sharing a single bit line and accessed as a group. Alternatively, memory elements may be configured so that each element is individually accessible, e.g., a NOR memory array. NAND and NOR memory configurations are examples, and memory elements may be otherwise configured.

[0071] The semiconductor memory elements located within and/or over a substrate may be arranged in two or three dimensions, such as a two-dimensional memory structure or a three-dimensional memory structure.

[0072] In a two-dimensional memory structure, the semiconductor memory elements are arranged in a single plane or a single memory device level. Typically, in a two-dimensional memory structure, memory elements are arranged in a plane (e.g., in an x-z direction plane) which extends substantially parallel to a major surface of a substrate that supports the memory elements. The substrate may be a wafer over or in which the layer of the memory elements are formed or it may be a carrier substrate which is attached to the memory elements after they are formed. As a non-limiting example, the substrate may include a semiconductor such as silicon.

[0073] The memory elements may be arranged in the single memory device level in an ordered array, such as in a plurality of rows and/or columns. However, the memory elements may be

arrayed in non-regular or non-orthogonal configurations. The memory elements may each have two or more electrodes or contact lines, such as bit lines and wordlines.

[0074] A three-dimensional memory array is arranged so that memory elements occupy multiple planes or multiple memory device levels, thereby forming a structure in three dimensions (i.e., in the x, y and z directions, where the y direction is substantially perpendicular and the x and z directions are substantially parallel to the major surface of the substrate).

[0075] As a non-limiting example, a three-dimensional memory structure may be vertically arranged as a stack of multiple two-dimensional memory device levels. As another non-limiting example, a three-dimensional memory array may be arranged as multiple vertical columns (e.g., columns extending substantially perpendicular to the major surface of the substrate, i.e., in the y direction) with each column having multiple memory elements in each column. The columns may be arranged in a two-dimensional configuration, e.g., in an x-z plane, resulting in a three-dimensional arrangement of memory elements with elements on multiple vertically stacked memory planes. Other configurations of memory elements in three dimensions can also constitute a three-dimensional memory array.

[0076] By way of non-limiting example, in a three-dimensional NAND memory array, the memory elements may be coupled together to form a NAND string within a single horizontal (e.g., x-z) memory device levels. Alternatively, the memory elements may be coupled together to form a vertical NAND string that traverses across multiple horizontal memory device levels. Other three-dimensional configurations can be envisioned wherein some NAND strings contain memory elements in a single memory level while other strings contain memory elements which span through multiple memory levels. Three-dimensional memory arrays may also be designed in a NOR configuration and in a ReRAM configuration.

[0077] Typically, in a monolithic three-dimensional memory array, one or more memory device levels are formed above a single substrate. Optionally, the monolithic three-dimensional memory array may also have one or more memory layers at least partially within the single substrate. As a non-limiting example, the substrate may include a semiconductor such as silicon. In a monolithic three-dimensional array, the layers constituting each memory device level of the array are typically formed on the layers of the underlying memory device levels of the array. However, layers of adjacent memory device levels of a monolithic three-dimensional memory array may be shared or have intervening layers between memory device levels.

[0078] Then again, two dimensional arrays may be formed separately and then packaged together to form a non-monolithic memory device having multiple layers of memory. For example, non-monolithic stacked memories can be constructed by forming memory levels on separate substrates and then stacking the memory levels atop each other. The substrates may be thinned or removed from the memory device levels before stacking, but as the memory device levels are initially formed over separate substrates, the resulting memory arrays are not monolithic three-dimensional memory arrays. Further, multiple two-dimensional memory arrays or three-dimensional memory arrays (monolithic or non-monolithic) may be formed on separate chips and then packaged together to form a stacked-chip memory device.

[0079] Associated circuitry is typically required for operation of the memory elements and for communication with the memory elements. As non-limiting examples, memory devices may have circuitry used for controlling and driving memory elements to accomplish functions such as programming and reading. This associated circuitry may be on the same substrate as the memory elements and/or on a separate substrate. For example, a controller for memory read-write operations may be located on a separate controller chip and/or on the same substrate as the memory elements.

[0080] One of skill in the art will recognize that this invention is not limited to the two dimensional and three-dimensional structures described but cover all relevant memory structures within the spirit and scope of the invention as described herein and as understood by one of skill in the art.

[0081] It is intended that the foregoing detailed description be understood as an illustration of selected forms that the invention can take and not as a definition of the invention. It is only the following claims, including all equivalents, that are intended to define the scope of the claimed invention. Finally, it should be noted that any aspect of any of the embodiments described herein can be used alone or in combination with one another.

Claims

1. A data storage device comprising: a non-volatile memory; and one or more processors, individually or in combination, configured to: detect that a data consumption rate of a host is below a threshold; and in response to detecting that the data consumption rate of the host is below the threshold, enter a power-saving mode by: storing, in the non-volatile memory, a context of a pending host command in a command queue; and reducing power to at least one component of the data storage device.
2. The data storage device of claim 1, wherein the one or more processors, individually or in combination, are further configured to exit the power-saving mode by: reading, from the non-volatile memory, the context of the pending host command; and restoring power to the at least one component of the data storage device.
3. The data storage device of claim 2, wherein the one or more processors, individually or in combination, are further configured to exit the power-saving mode in response to an expiration of a time period.
4. The data storage device of claim 3, wherein the time period is a function of the data consumption rate of the host and a size of a buffer in the host.
5. The data storage device of claim 2, wherein the one or more processors, individually or in combination, are further configured to exit the power-saving mode in response to receiving a new host command from the host.
6. The data storage device of claim 2, wherein the one or more processors, individually or in combination, are further configured to exit the power-saving mode in response to receiving a command from the host to exit the power-saving mode.
7. The data storage device of claim 1, wherein the one or more processors, individually or in combination, are further configured to seek approval from the host of an increase in latency in exiting the power-saving mode.
8. The data storage device of claim 1, wherein the one or more processors, individually or in combination, are further configured to request that the host not send a new host command to the data storage device while the data storage device is in the power-saving mode.
9. The data storage device of claim 1, wherein the one or more processors, individually or in combination, are further configured to perform a handshake with the host to enable use of the power-saving mode.
10. The data storage device of claim 1, wherein the non-volatile memory comprises a three-dimensional memory.
11. A method comprising: performing in a data storage device: determining that there is at least one outstanding host command that has not been completed; and entering a low-power mode even though there is the at least one outstanding host command that has not been completed.
12. The method of claim 11, wherein entering the low-power mode comprises: storing a state of the at least one outstanding host command; and reducing power to at least one component of the data storage device.
13. The method of claim 11, further comprising exiting the low-power mode.
14. The method of claim 13, wherein exiting the low-power mode comprises: restoring the state of the at least one outstanding host command; and restoring power to the at least one component of the data storage device.

- 15.** The method of claim 13, wherein the low-power mode is exited in response to an expiration of a time period, receiving a new host command from the host, and/or receiving a command from the host to exit the low-power mode.
 - 16.** The method of claim 11, further comprising: receiving consent from the host for an increase in latency in exiting the low-power mode.
 - 17.** The method of claim 11, further comprising: requesting that the host not send a new host command to the data storage device while the data storage device is in the low-power mode.
 - 18.** The method of claim 11, further comprising: performing a handshake with the host to enable use of the low-power mode.
 - 19.** The method of claim 11, wherein the host comprises an autonomous vehicle.
 - 20.** A data storage device comprising: a memory; and means for enter a standby mode in response to detecting that a data consumption rate of a host is below a threshold, wherein the standby mode power is entered even though there is at least one pending host command.
-